AutoDD: Learning to Select Drift Detectors for Evolving Data Streams

Anonymous

***Abstract*—Concept drift, the change in the distribution of streaming data over time, significantly impacts the performance of machine learning models. While numerous drift detection methods have been proposed, selecting and configuring the most suitable detector for a specific data stream remains a challenging task. Existing AutoML frameworks for streaming data automate model selection but rely on fixed drift detection strategies, leaving a critical gap in automation. This paper introduces AutoDD (Automated Drift Detector), a novel framework that jointly automates detector selection and hyperparameter configuration using a hierarchical learning approach. AutoDD extracts comprehensive online meta-features and employs a two-level hierarchical selection algorithm to adaptively choose both the detector and its configuration. Under standard contextual-bandit assumptions, our analysis gives sublinear regret bounds for the proxy selection objective optimized by AutoDD. Comprehensive experiments on synthetic and real-world datasets demonstrate that AutoDD achieves strong mean performance relative to state-of-the-art detectors while maintaining computational efficiency.**

I. Introduction

Concept drift, where the statistical properties of streaming data evolve over time, poses a significant challenge in data stream mining [1], incremental learning [2], and non-stationary learning [3]. As data distributions change, machine learning models experience degraded performance, necessitating timely detection and adaptation. This phenomenon manifests in various forms, including sudden (abrupt) drift, gradual drift, incremental drift, and recurring drift, as illustrated in Figure 1. For instance, an electricity price prediction model may encounter sudden drift when market regulations change abruptly, experience gradual drift as consumer behavior shifts over seasons, or observe recurring drift driven by daily demand cycles [3].

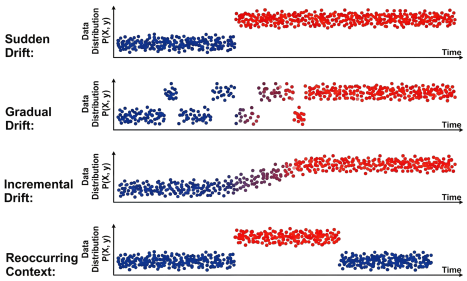


Fig. 1: Types of concept drifts.

Over the past two decades, numerous drift detection methods have been developed to address this challenge, including error rate-based approaches such as DDM [4], ADWIN [5], and ADDM [6], distribution-based methods like KSWIN [7], uncertainty-based methods such as PUDD [8], and meta-learning techniques like Meta-ADD [9]. However, a fundamental challenge remains unresolved in systematically selecting and configuring the most appropriate drift detector for a given data stream.

Automated Machine Learning (AutoML) has revolutionized the field by addressing similar selection challenges in batch learning scenarios [10]. Recent advancements have extended AutoML to streaming data, enabling automatic selection and configuration of classifiers [11], [12]. However, these systems treat drift detection as a fixed component, requiring manual specification and leaving a critical gap in the automation framework.

To bridge this gap, we introduce **AutoDD** (**A**utomated **D**rift **D**etector). This modular framework automates drift detector selection and configuration using a hierarchical learning approach, compatible with any model or AutoML pipeline.

- We propose a two-level hierarchical selection algorithm using LinUCB [13] and Thompson Sampling [14] that adaptively selects detectors and their hyperparameters based on online meta-features.
- We formalize detector selection with a theoretical analysis that provides sublinear regret bounds for the proxy objective used by the selection layer.
- Extensive evaluation demonstrates AutoDD achieves higher mean accuracy than the compared state-of-the-art detectors while maintaining computational efficiency.

The rest of this paper is organized as follows. Section II presents the related work. Section III provides the details of our proposed AutoDD framework. Section IV describes the experimental setup. Section V presents the experimental results and analysis. Finally, Section VII concludes the paper.

II. Related Work

A. Data Streams and Concept Drift

A data stream is a continuous, potentially unbounded sequence of instances $\{(\mathbf{x}_t, y_t)\}_{t=1}^{\infty}$ that arrive over time, often at high velocity, from sources such as sensor networks, network-traffic monitors, financial tickers, and web and social-media activity [1], [2]. Unlike a static dataset that can be stored and revisited, a stream must be processed online: instances may arrive faster than they can be persisted, and the cumulative volume can exceed the available memory. A stream learner is therefore expected to process each instance at most once, use bounded time and memory per instance, remain ready to predict at any point, and adapt to temporal changes in the

data [15]. These constraints rule out the multi-pass, full-data access assumed by batch learning and offline AutoML.

The central difficulty of this setting is that the data-generating process is non-stationary, so that its statistical properties evolve over time, a phenomenon termed concept drift [1], [3] and formalized as $P_t(\mathbf{x}, y) \neq P_{t'}(\mathbf{x}, y)$ for some $t \neq t'$. As illustrated in Figure 1, drift is commonly categorized by how the change unfolds over time into sudden, gradual, incremental, and recurring drift [3]. Because a detector or model tuned for one regime degrades once the stream moves to another, timely drift detection and subsequent adaptation are essential, which motivates both the detection methods surveyed next and the automated selection we propose.

B. Drift Detection and Streaming AutoML

Drift detection methods can be broadly categorized into three paradigms. **Error rate-based methods** monitor fluctuations in classifier error rates, triggering alarms when changes are statistically significant. DDM [4] pioneered the warning-drift framework by monitoring prediction accuracy under binomial distribution assumptions, followed by EDDM [16] which focuses on distances between errors to detect gradual drift, HDDM [17] which applies Hoeffding's inequality with exponentially weighted moving averages, and ADWIN [5] which maintains adaptive sliding windows that automatically shrink when distribution changes are detected. Page-Hinkley [18] applies cumulative sum control charts to detect mean shifts in sequential data. Recent advances include MDDM [19], which uses McDiarmid's inequality for shorter detection delays with tighter concentration bounds, and ADDM [6], which models error rates as SETAR time series [20] to achieve lower false alarm rates. **Distribution-based methods** compare data distributions across windows by quantifying dissimilarity between new and historical data distributions [21], including approaches based on information theory [22], competence models [23], and the Kolmogorov-Smirnov test (KSWIN) [7] which detects changes in univariate feature distributions. Fully unsupervised, model-independent variants such as the image-based detector IBDD [24] render high-dimensional windows as grayscale images and monitor their mean squared deviation, enabling label-free detection on fast, high-dimensional streams. **Uncertainty and learning-based methods** address limitations of error-rate monitoring: PUDD [8] leverages prediction confidence for earlier detection by monitoring changes in prediction uncertainty distributions, while Meta-ADD [9] employs prototypical networks to classify drift types via meta-learning on diverse drift scenarios.

Recent developments in automated algorithm selection and hyperparameter optimization (CASH) [10] have been extended to streaming scenarios [25]. However, all existing streaming AutoML systems assume a pre-selected drift detector, requiring manual specification and leaving a critical gap in the automation framework. Despite advances in both drift detection and streaming AutoML, drift detector selection remains manual. Our work fills this gap by extending AutoML to automatic drift detector selection and configuration.

TABLE I: Meta-feature portfolio with mathematical formulations.

Category	Feature	Formula/Description
Statistical	Velocity	$v_t = n_t/\Delta t$
	Dimensionality	d
	Sparsity	$\frac{1}{wd}\sum_{i=1}^{w}\sum_{j=1}^{d}\mathbb{I}[\mathbf{x}_{i,j}=0]$
	Noise level	$H(\mathbf{X}) \approx -\frac{1}{w}\sum_{i=1}^{w}\log \hat{p}_k(\mathbf{x}_i)$ (k-NN, $k=5$)
	Class imbalance	$\max_c \lvert C_c\rvert / \min_c \lvert C_c\rvert$
Stationarity	Error volatility	$\sigma_{\text{err}}(t) = \sqrt{\frac{1}{w}\sum_{i=t-w}^{t}(e_i - \bar{e})^2}$
	KL divergence	$D_{\text{KL}}(P_t \Vert P_{t-1})$ (histogram, $\sqrt{w}$ bins)
	Autocorrelation	$\rho_t = \frac{\text{Cov}(e_t, e_{t-1})}{\sigma_{e_t}\sigma_{e_{t-1}}}$
	Mean shift	$\Delta\mu_t = \frac{1}{d}\sum_{j=1}^{d}\lvert\mu_{t,j} - \mu_{t-1,j}\rvert$
	Variance shift	$\Delta\sigma_t^2 = \frac{1}{d}\sum_{j=1}^{d}\lvert\sigma_{t,j}^2 - \sigma_{t-1,j}^2\rvert$
Temporal	Trend	$\beta_{\text{trend}} = \frac{\text{Cov}(t, e_t)}{\text{Var}(t)}$
	Periodicity	$f_{\text{dom}} = \arg\max_f \lvert\mathcal{F}\{e_t\}(f)\rvert$
	Change frequency	$\frac{1}{\Delta t}\sum_{i=1}^{w}\mathbb{I}[\lvert\Delta e_i\rvert > \tau_{\Delta e}]$
	Change magnitude	$\frac{1}{\lvert C_t\rvert}\sum_{i\in C_t}\lvert\Delta e_i\rvert$
	Recurrence score	$\frac{1}{w^2}\sum_{i,j}\mathbb{I}[\lVert\mathbf{x}_i - \mathbf{x}_j\rVert < \epsilon]$

III. Automated Drift Detector

AutoDD employs an adaptive, closed-loop architecture to dynamically select and configure drift detectors. As shown in Figure 2, this architecture operates through several key steps. First, the system initializes the detector portfolio with standard hyperparameters and sets up the learning parameters. During online operation, meta-features characterizing stream velocity, stationarity, and temporal patterns are continuously extracted from the incoming data stream. These meta-features serve as context for the LinUCB agent, which selects a detector from the portfolio. Thompson Sampling then chooses a hyperparameter configuration for the selected detector. The configured detector monitors the stream and triggers model adaptation upon detecting drift. Finally, performance feedback is computed based on detection outcomes, which updates both the LinUCB and Thompson Sampling components to refine future selections. This adaptive learning cycle enables AutoDD to improve its detector selections over time.

A. Algorithm Overview

AutoDD operates over sliding windows of size w. When each window fills with incoming instances, the framework executes a complete selection cycle, beginning with meta-feature extraction.

Meta-Feature Extraction: The meta-feature set comprises 15 features organized into three categories: Statistical, Stationarity, and Temporal. Table I summarizes each meta-feature with its mathematical formulation. All features are computed over a sliding window of size w using online statistics and normalized to $[0, 1]$ via running min-max normalization to ensure scale consistency. The resulting context vector $\boldsymbol{\phi}_t \in \mathbb{R}^m$ with $m = 15$ is computed incrementally, supporting efficient per-instance updates.

Detector Portfolio: AutoDD maintains a diverse portfolio of $K = 7$ drift detectors spanning window-based, error-based, and distribution-based paradigms. Table II summarizes each detector with its hyperparameters, valid ranges, sampling scales, and default values from the literature. For each detector i, we maintain C hyperparameter configurations denoted $\boldsymbol{\eta}_{i,c}$ for $c \in [C]$, the first is always the literature default, while the

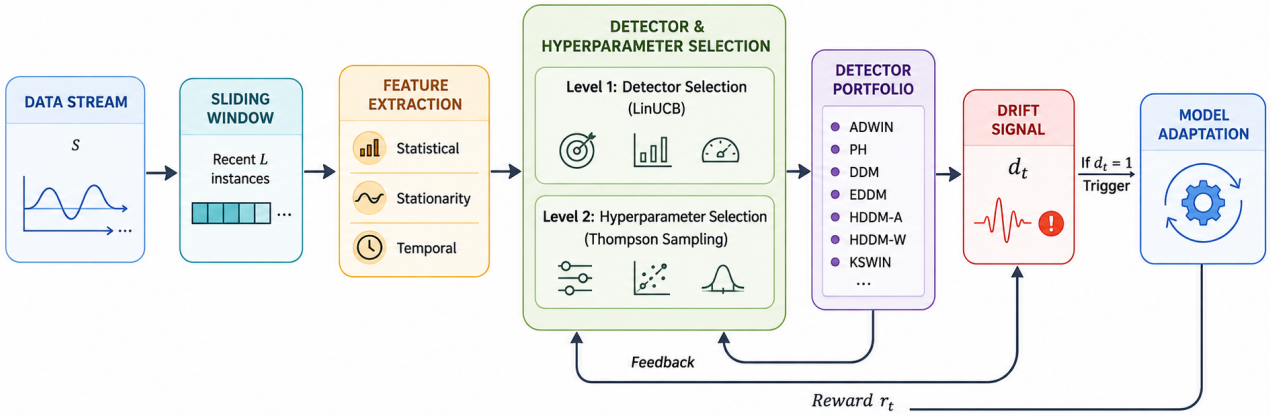


Fig. 2: The AutoDD framework: meta-feature extraction, hierarchical selection via LinUCB and Thompson Sampling, and detector portfolio with feedback loop.

TABLE II: Detector portfolio with hyperparameter configurations.

Detector	Type	Parameter	Range	Scale	Default
ADWIN	Window-based	δ_{adwin}	[0.0001, 0.1]	log	0.002
Page-Hinkley	Window-based	θ_{ph}	[10, 100]	int	50
		δ_{ph}	[0.001, 0.01]	log	0.005
DDM	Error-based	w_{thresh}	[1.5, 3.0]	linear	2.0
		d_{thresh}	[2.0, 4.0]	linear	3.0
EDDM	Error-based	α_{eddm}	[0.90, 0.99]	linear	0.95
		β_{eddm}	[0.85, 0.95]	linear	0.90
HDDM-A	Error-based	δ_{hddm}	[0.0001, 0.01]	log	0.001
HDDM-W	Error-based	λ_{hddm}	[0.01, 0.1]	log	0.05
KSWIN	Distribution-based	α_{ks}	[0.001, 0.05]	log	0.005
		w_{ks}	[50, 500]	int	100

remaining $C-1$ are sampled from the specified ranges using the appropriate logarithmic, linear, or integer scale.

B. Hierarchical Learning Approach

AutoDD uses a two-level hierarchical selection strategy that jointly adapts detector choice and hyperparameter configuration while maintaining computational efficiency. This decomposition avoids directly treating all $K \times C$ detector-configuration pairs as a single flat action space.

Level 1 - Detector Selection via LinUCB: We formulate detector selection as a contextual multi-armed bandit problem where context is $\boldsymbol{\phi}_t$ (meta-features), arms are K detectors, and reward is detection performance. LinUCB [13] assumes linear reward:

$$r_{i,t} = \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_i + \epsilon_t \tag{1}$$

where $\boldsymbol{\theta}_i \in \mathbb{R}^m$ is the unknown parameter vector for detector i, with m being the meta-feature dimensionality. The algorithm selects the detector with the highest upper confidence bound:

$$i_t = \arg\max_{i \in [K]} \left(\boldsymbol{\phi}_t^\top \hat{\boldsymbol{\theta}}_i + \alpha_{\text{ucb}} \sqrt{\boldsymbol{\phi}_t^\top \mathbf{A}_i^{-1} \boldsymbol{\phi}_t} \right) \tag{2}$$

where $\hat{\boldsymbol{\theta}}_i = \mathbf{A}_i^{-1}\mathbf{b}_i$ is the ridge regression estimate. Here, $\mathbf{A}_i = \lambda_{\text{reg}}\mathbf{I} + \sum_{s:i_s=i} \boldsymbol{\phi}_s \boldsymbol{\phi}_s^\top \in \mathbb{R}^{m \times m}$ is the regularized design matrix with regularization parameter $\lambda_{\text{reg}} > 0$, and $\mathbf{b}_i = \sum_{s:i_s=i} r_s \boldsymbol{\phi}_s \in \mathbb{R}^m$ is the cumulative reward-weighted context vector for detector i. The parameter $\alpha_{\text{ucb}} > 0$ controls exploration-exploitation balance.

Level 2 - Hyperparameter Selection via Thompson Sampling: Given the selected detector D_{i_t}, we adaptively configure its hyperparameters using Thompson Sampling [14]. For each configuration $c \in [C] = \{1, \ldots, C\}$ of detector i, we maintain Beta distribution parameters $(a_{i,c}, b_{i,c})$ initialized to $(1, 1)$. At each window, we sample:

$$\psi_{i_t,c} \sim \text{Beta}(a_{i_t,c}, b_{i_t,c}) \quad \forall c \in [C] \tag{3}$$

and select configuration:

$$c_t = \arg\max_c \psi_{i_t,c} \tag{4}$$

After observing reward r_t, we update:

$$a_{i_t,c_t} \leftarrow a_{i_t,c_t} + \tilde{r}_t \tag{5}$$

$$b_{i_t,c_t} \leftarrow b_{i_t,c_t} + (1 - \tilde{r}_t) \tag{6}$$

where $\tilde{r}_t = (r_t + 1)/2$ normalizes the reward to $[0, 1]$.

C. Evaluation and Adaptation

Evaluating detector performance without ground-truth drift labels is challenging. Following standard practice, we use model accuracy drop as a proxy metric. The reward at time t is:

$$r_t = \begin{cases} 0.5 + \min(\Delta a_t \cdot 2, 0.5) & d_t = 1, \Delta a_t > \tau_{\text{sig}} \\ -0.3 & d_t = 1, \Delta a_t \leq \tau_{\text{sig}} \\ -0.5 - \min(\Delta a_t, 0.5) & d_t = 0, \Delta a_t > \tau_{\text{sig}} \\ 0.1 & d_t = 0, \Delta a_t \leq \tau_{\text{sig}} \end{cases} \tag{7}$$

where $d_t \in \{0, 1\}$ indicates drift detection at time t, $\Delta a_t = a_{t-w} - a_t$ is the accuracy drop between consecutive windows, and $\tau_{\text{sig}} = 0.05$ is the significance threshold. The reward function penalizes both false positives and false negatives

while rewarding correct drift detections and stable periods. We note that an accuracy-based proxy couples detector quality with the downstream learner, label noise, and the evaluation window. Two design choices limit the resulting variance: the drop Δa_t is measured over a window of w instances rather than per-instance, smoothing transient noise, and the threshold τ_{sig} suppresses rewards from sub-significant fluctuations so that noise-driven dips do not reinforce spurious detections. Because the reward enters the bandit update only as a bounded scalar in $[-1, 1]$, occasional misattributed rewards perturb the posteriors gradually rather than abruptly. The regret analysis in Section III is therefore stated for this bounded proxy reward; interpreting proxy regret as true detector quality requires the proxy-fidelity assumption introduced below.

Upon drift detection, the model undergoes a complete reset. Let $\mathcal{M}_t$ denote the model state at time t. When drift is detected ($d_t = 1$), we perform:

$$\mathcal{M}_t \leftarrow \mathcal{M}_0 \tag{8}$$

where $\mathcal{M}_0$ represents the initial untrained state. The model then resumes incremental learning from subsequent instances:

$$\mathcal{M}_{t+s} \leftarrow \text{Update}(\mathcal{M}_{t+s-1}, (\mathbf{x}_{t+s}, y_{t+s})) \quad \forall s > 0 \tag{9}$$

The complete procedure is formalized in Algorithm 1.

Algorithm 1: AutoDD

Input: Stream $\mathcal{S}$, portfolio $\mathcal{D}$, window w; exploration α_{ucb}, regularization λ_{reg}, configs per detector C
Output: Drift signals $\{d_t\}_{t=1}^{\infty}$

Initialize $\mathbf{A}_i \leftarrow \lambda_{\text{reg}}\mathbf{I}_m$, $\mathbf{b}_i \leftarrow \mathbf{0}$ for all $i \in [K]$;
Initialize $a_{i,c} \leftarrow 1$, $b_{i,c} \leftarrow 1$ for all $i \in [K], c \in [C]$;
Initialize model $\mathcal{M}_0$, sliding window $W \leftarrow \emptyset$;
foreach *instance* $(\mathbf{x}_t, y_t) \in \mathcal{S}$ **do**
  $W \leftarrow W \cup \{(\mathbf{x}_t, y_t)\}$;
  if $|W| = w$ **then**
    $\boldsymbol{\phi}_t \leftarrow \text{EXTRACT}(W)$; `// meta-features`
    for $i = 1$ **to** K **do**
      $\text{UCB}_i \leftarrow \boldsymbol{\phi}_t^\top \mathbf{A}_i^{-1}\mathbf{b}_i + \alpha_{\text{ucb}}\sqrt{\boldsymbol{\phi}_t^\top \mathbf{A}_i^{-1}\boldsymbol{\phi}_t}$;
    $i_t \leftarrow \arg\max_i \text{UCB}_i$; `// select detector`
    for $c = 1$ **to** C **do**
      Sample $\psi_c \sim \text{Beta}(a_{i_t,c}, b_{i_t,c})$;
    $c_t \leftarrow \arg\max_c \psi_c$; `// select configuration`
    $d_t \leftarrow D_{i_t}(\boldsymbol{\eta}_{i_t,c_t}).\text{DETECT}(W)$;
    if $d_t = 1$ **then**
      $\mathcal{M}_t \leftarrow \mathcal{M}_0$; `// reset model`
    Compute reward r_t using Eq. (7);
    $\mathbf{A}_{i_t} \leftarrow \mathbf{A}_{i_t} + \boldsymbol{\phi}_t\boldsymbol{\phi}_t^\top$; $\mathbf{b}_{i_t} \leftarrow \mathbf{b}_{i_t} + r_t\boldsymbol{\phi}_t$;
    $\tilde{r}_t \leftarrow (r_t + 1)/2$;
    $a_{i_t,c_t} \leftarrow a_{i_t,c_t} + \tilde{r}_t$; $b_{i_t,c_t} \leftarrow b_{i_t,c_t} + (1 - \tilde{r}_t)$;
    $W \leftarrow \emptyset$;

D. Computational Complexity

AutoDD has per-window complexity of $O(wd + Km^2)$ for the incrementally maintained meta-features used in our implementation, where w is the AutoDD window size, d is the input feature dimensionality, K is the number of detectors, and $m = 15$ is the meta-feature count. The main costs are meta-feature extraction ($O(wd)$) and LinUCB matrix operations ($O(Km^2)$). Space complexity is $O(Km^2 + wd)$ for storing K covariance matrices of size $m \times m$ and the sliding window buffer.

E. Theoretical Analysis

We formalize the optimization objective as a streaming detector selection problem and provide proxy-regret bounds for the hierarchical learning approach. Our analysis focuses on the selection mechanism, showing that the average proxy suboptimality vanishes under the stated contextual-bandit assumptions.

Problem Formulation. A data stream $\mathcal{S} = \{(\mathbf{x}_t, y_t)\}_{t=1}^{\infty}$ exhibits concept drift when distributions change over time:

$$\exists t, t' : P_t(\mathbf{x}, y) \neq P_{t'}(\mathbf{x}, y) \tag{10}$$

Given a detector portfolio $\mathcal{D} = \{D_1, \ldots, D_K\}$ where each D_i has configuration space $\mathcal{H}_i$, AutoDD addresses the streaming detector selection and configuration problem. Inspired by the CASH formulation [10], we jointly optimize the detector choice and its hyperparameters:

$$(D_t^*, \boldsymbol{\eta}_t^*) = \arg\max_{D_i \in \mathcal{D}, \boldsymbol{\eta} \in \mathcal{H}_i} \mathbb{E}[\mathcal{M}(D_i(\boldsymbol{\eta}), \mathcal{S}_t) \mid \boldsymbol{\phi}_t] \tag{11}$$

where $\boldsymbol{\phi}_t$ is the stream context, $\mathcal{S}_t$ denotes the stream segment at time t, $\mathbb{E}[\cdot]$ is the expectation operator, and $\mathcal{M}$ denotes the downstream model performance.

Assumptions. We adopt three standard assumptions from the contextual bandit literature [26], [27], together with a fidelity condition (A4) that links our accuracy-based proxy reward to true detection quality:

- **(A1) Bounded features:** $\|\boldsymbol{\phi}_t\|_2 \leq L$ for all t, satisfied by $[0, 1]$ normalization with $L = \sqrt{m}$.
- **(A2) Linear reward:** For each detector i, there exists $\boldsymbol{\theta}_i^* \in \mathbb{R}^m$ with $\|\boldsymbol{\theta}_i^*\|_2 \leq S$ such that $\mathbb{E}[r_{i,t} \mid \boldsymbol{\phi}_t] = \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_i^*$, where $S > 0$ is a constant upper bound on the parameter norms.
- **(A3) Sub-Gaussian noise:** Reward noise is σ_{noise}-sub-Gaussian, satisfied with $\sigma_{\text{noise}} = 1$ for bounded rewards in $[-1, 1]$.
- **(A4) Proxy fidelity:** The windowed accuracy-drop reward (Eq. (7)) is, in expectation, monotone in true detection quality, so that a high proxy-reward detector-configuration pair is also high quality under the objective in Eq. (11). The bounds below are stated with respect to this proxy objective; reading proxy regret as evidence of convergence toward a high-quality detector additionally relies on A4 as a modeling assumption.

These assumptions characterize the *selection layer* under the standard linear contextual-bandit abstraction; they are not claims about the full loop spanning detection, adaptation, and learning. A2 in particular treats the windowed reward as linear in the meta-features in expectation, an approximation we do not prove directly but examine empirically through the selection-accuracy trend in Table VIII.

Remark (stationarity under drift). Although the data distribution P_t is non-stationary by construction, A2 requires only that the *conditional* reward map $\boldsymbol{\phi}_t \mapsto \mathbb{E}[r_{i,t} \mid \boldsymbol{\phi}_t]$ be stationary, not the marginal P_t. This is the intended role of the meta-features: $\boldsymbol{\phi}_t$ is designed to encode the current stream regime (velocity, stationarity, and temporal structure), so changes in P_t can be represented through the context rather than through time-varying $\boldsymbol{\theta}_i^*$. A detector well matched to a recurring regime should earn comparable reward whenever that regime recurs, keeping $\boldsymbol{\theta}_i^*$ approximately fixed across time. When this abstraction is imperfect, i.e., the reward map itself drifts, sliding-window or discounted LinUCB variants could instead be analyzed using dynamic-regret tools with a variation budget $B_T = \sum_t \|\boldsymbol{\theta}_{i,t}^* - \boldsymbol{\theta}_{i,t-1}^*\|_2$ [27]. The monotonic rise of proxy-oracle selection accuracy in Table VIII is consistent with the stationary approximation on the streams we study.

The following result gives the detector-selection proxy regret for LinUCB under the stationary linear abstraction.

Theorem 1 (Level 1 Regret). Under assumptions A1–A3, with exploration parameter $\alpha_{\text{ucb}} = \sigma_{\text{noise}}\sqrt{m\log(KT/\delta_{\text{conf}})} + S$ where T is the time horizon and $\delta_{\text{conf}} \in (0,1)$ is the confidence parameter, LinUCB detector selection achieves with probability at least $1-\delta_{\text{conf}}$:

$$R_1(T) = \sum_{t=1}^{T}\left(\max_i \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_i^* - \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_{i_t}^*\right) = O(\sqrt{KmT\log T}) \tag{12}$$

Proof. The proof follows the disjoint-arm LinUCB analysis of [26]. By the self-normalized martingale inequality (Theorem 1, [26]), the true parameter $\boldsymbol{\theta}_i^*$ lies within the confidence ellipsoid $\mathcal{C}_{i,t} = \{\boldsymbol{\theta} : \|\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}_i\|_{\mathbf{A}_i} \le \alpha_{\text{ucb}}\}$ with probability $1-\delta_{\text{conf}}$. Under this event, UCB selection ensures $\boldsymbol{\phi}_t^\top \boldsymbol{\theta}_{i_t}^* \ge \max_i \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_i^* - 2\alpha_{\text{ucb}}\|\boldsymbol{\phi}_t\|_{\mathbf{A}_{i_t}^{-1}}$. Applying the elliptical potential lemma [28] separately to each detector estimator and then summing over detectors gives the additional $\sqrt{K}$ factor, yielding $R_1(T) = O(\sqrt{KmT\log T})$ under the standard convention that treats the confidence radius α_{ucb} as constant. The full three-step derivation, including the dimension-explicit form $\tilde{O}(m\sqrt{KT})$ obtained by retaining α_{ucb}, is given in Appendix A. □

Thompson Sampling subsequently balances exploration and exploitation for hyperparameter configuration.

Proposition 2 (Level 2 Proxy Regret). For Thompson Sampling on C configurations per detector under the standard Bernoulli-sampling reduction for bounded rewards, with $N_i(T)$ denoting the number of times detector $i \in [K]$ is selected over T rounds, the expected cumulative proxy regret satisfies:

$$R_2(T) = \sum_{i=1}^{K} O(\sqrt{C \cdot N_i(T)\log N_i(T)}) \tag{13}$$

Proof. For each detector $i \in [K]$, conditioned on detector i being selected, hyperparameter selection reduces to a C-armed bandit with bounded rewards. We normalize $r_t \in [-1,1]$ to $\tilde{r}_t = (r_t+1)/2 \in [0,1]$. The standard reduction draws a Bernoulli outcome with mean $\tilde{r}_t$ and updates the Beta posterior with that sampled outcome; by Theorem 1 of [29], Thompson Sampling achieves expected regret $O(\sqrt{CN_i(T)\log N_i(T)})$ over $N_i(T)$ rounds. Since $\sum_i N_i(T) = T$, summing across all detectors yields the stated bound. The fractional update in Eq. (5) is the conditional expectation of this reduction and is used as a stable implementation heuristic rather than a separate regret theorem. The full reduction-and-aggregation argument is given in Appendix A. □

Under the same proxy-reward abstraction, the combined regret remains sublinear, implying that average per-round proxy suboptimality vanishes over time.

Corollary (Combined Proxy Regret). Let (i_t^*, c_t^*) denote the per-context proxy-oracle detector-configuration pair at round t. If the hierarchical proxy objective decomposes into detector-selection and within-detector configuration terms, then the total regret can be written as $R(T) = R_1(T) + R_2(T)$, where $R_1(T)$ captures detector-selection suboptimality and $R_2(T)$ captures configuration-selection suboptimality. The combined bound is:

$$R(T) = O(\sqrt{KmT\log T} + \sqrt{CKT\log T}) \tag{14}$$

Proof. Under the decomposition assumption, the per-round gap can be telescoped through an intermediate term: writing $c_{i_t}^*$ for the best configuration of the *selected* detector i_t, the gap between the proxy-oracle pair (i_t^*, c_t^*) and the played pair (i_t, c_t) splits into detector-selection suboptimality and configuration-selection suboptimality. The first term is bounded by Theorem 1 and the second by Proposition 2. Summing over t and applying the Cauchy-Schwarz inequality, $\sum_i \sqrt{N_i(T)} \le \sqrt{K\sum_i N_i(T)} = \sqrt{KT}$, gives $R_2(T) = O(\sqrt{CKT\log T})$, which combined with $R_1(T)$ yields the result. The detailed combination is given in Appendix A. □

Thus, under the proxy-reward and decomposition assumptions, the average proxy regret vanishes:

$$\lim_{T\to\infty} \frac{R(T)}{T} = 0 \tag{15}$$

Empirical support for this learning behavior is provided in Table VIII, where AutoDD's proxy-oracle selection accuracy improves from 69.4% in the first quarter of the stream to 93.1% by stream end. This trend is consistent with the linear reward approximation on the evaluated streams, but does not by itself prove the approximation. As a concrete reference point, inverting the dominant $O(\sqrt{KmT\log T})$ term for our portfolio ($K = 7$, $m = 15$) to reach an average proxy suboptimality on the order of 10^{-2} requires a horizon of roughly $T \approx 5 \times 10^5$ windows, which matches the order of magnitude over which the convergence in Table VIII is observed.

TABLE III: Dataset characteristics (Multiple* = Sudden, Gradual, Incremental, and Normal drift types)

Dataset	Type	Instances	Features	Drift Type
SEA	Synthetic	500,000	3	Multiple*
HYP	Synthetic	500,000	10	Multiple*
AGR	Synthetic	100,000	9	Multiple*
RBF	Synthetic	200,000	10	Multiple*
RTG	Synthetic	100,000	30	Multiple*
Elec	Real-world	45,312	8	Unknown
Weather	Real-world	18,159	8	Unknown
Spam	Real-world	4,601	57	Unknown
Airlines	Real-world	539,383	7	Unknown
Poker	Real-world	829,201	10	Unknown

IV. Experiments

We evaluate AutoDD on ten benchmark datasets comprising five synthetic and five real-world streams, with characteristics detailed in Table III. To isolate drift detection performance from model selection effects, all methods use two fixed base models: GaussianNB and Hoeffding Adaptive Tree (HAT) [30]. These two learners represent fundamentally different model families (probabilistic and tree-based) and are the standard benchmark base learners used in the data stream mining literature [15], [31]. We compare AutoDD against twelve baselines: a no-detector baseline, seven portfolio detectors including ADWIN, DDM, EDDM, HDDM-A, HDDM-W, Page-Hinkley, KSWIN, and four state-of-the-art methods including MDDM [19], ADDM [6], PUDD [8], and Meta-ADD [9]. All detectors use recommended hyperparameter settings from the literature.

We measure prequential accuracy as the percentage of correct predictions using test-then-train evaluation [15]. For synthetic datasets with known drift points, we report F1-score computed by matching detected drifts to ground truth using a tolerance window of $\tau_{\text{tol}} = 100$ instances following standard practice [5]. For real-world datasets, we report the number of detected drifts (DN). These metrics are complementary by design, as prequential accuracy captures the downstream task impact of a detector's adaptation decisions, whereas the drift-detection F1-score scores precision and recall against ground-truth drift points and thus credits timely, correct detections while penalizing both missed drifts and spurious alarms that accuracy alone would not separate. On real-world streams the drift points are unknown, so F1 cannot be computed and we instead report DN alongside accuracy; following the benchmarking guidance of Souza et al. [32], we interpret DN relative to the consensus detection range across methods rather than as an absolute target, treating values far above or below this range as over- and under-detection respectively. AutoDD uses all 15 meta-features with window size $w = 100$, exploration parameter $\alpha_{\text{ucb}} = 1.0$, regularization parameter $\lambda_{\text{reg}} = 1.0$, and $C = 5$ hyperparameter configurations per detector. Results are averaged over 10 runs with random seeds uniformly sampled from the range [1, 100]. Implementation uses Python with River [33] on a MacBook Pro with Apple M2 Pro and 16GB RAM.[1]

V. Results

We conduct a comprehensive evaluation of AutoDD against classical detectors and state-of-the-art methods across both base models.

A. Synthetic Datasets

Table IV presents results on synthetic benchmarks with known drift points. AutoDD achieves the highest mean accuracy across both base models, indicating robust performance across the two learning paradigms evaluated. With GaussianNB, AutoDD attains 81.9% mean accuracy, while with HAT it reaches 84.0%, exceeding the compared state-of-the-art detectors in both settings. Meta-ADD ranks as the second-best method, achieving 80.9% with GaussianNB and 81.4% with HAT. AutoDD shows particular strength on datasets with abrupt drift patterns, where its ability to switch between detectors yields substantial benefits. On the SEA dataset, AutoDD achieves 93.1% accuracy with GaussianNB and 96.7% with HAT, outperforming all baselines in this setting. Similar improvements are observed on the AGR dataset, with 78.5% and 82.4% accuracy, respectively. Meta-ADD performs competitively on gradual drift scenarios due to its learned drift-type representations. Notably, fixed detectors exhibit inconsistent performance across different drift types, with HDDM-W achieving strong results on some datasets but underperforming on others, motivating adaptive detector selection. A detailed per-drift-type breakdown (sudden, incremental, gradual, recurring) is provided in Appendix A (Table IX).

B. Real-world Datasets

Table V presents results on real-world benchmarks where drift patterns are unknown and potentially more complex than synthetic scenarios. AutoDD achieves the highest mean accuracy with both base models. With GaussianNB, AutoDD attains 79.1% mean accuracy, improving to 83.1% with HAT, exceeding the compared state-of-the-art detectors on the mean metric. ADDM emerges as the second-best performer with 75.1% accuracy using GaussianNB and 77.8% with HAT. AutoDD's improvements appear across all five real-world datasets, suggesting that its adaptive selection mechanism transfers across the evaluated domains. On the Electricity dataset, AutoDD achieves 89.6% with GaussianNB and 93.8% with HAT. On Weather, it reaches 80.0% and 85.1% respectively, while on Spam it attains 97.9% and 99.0%. AutoDD also maintains a detection count within the consensus range of the compared methods, avoiding the most extreme over- and under-detection patterns. On the Airlines dataset, AutoDD detects 118 drifts, compared to HDDM-W's 269 detections and ADWIN's 76 detections, indicating more moderate sensitivity in this benchmark.

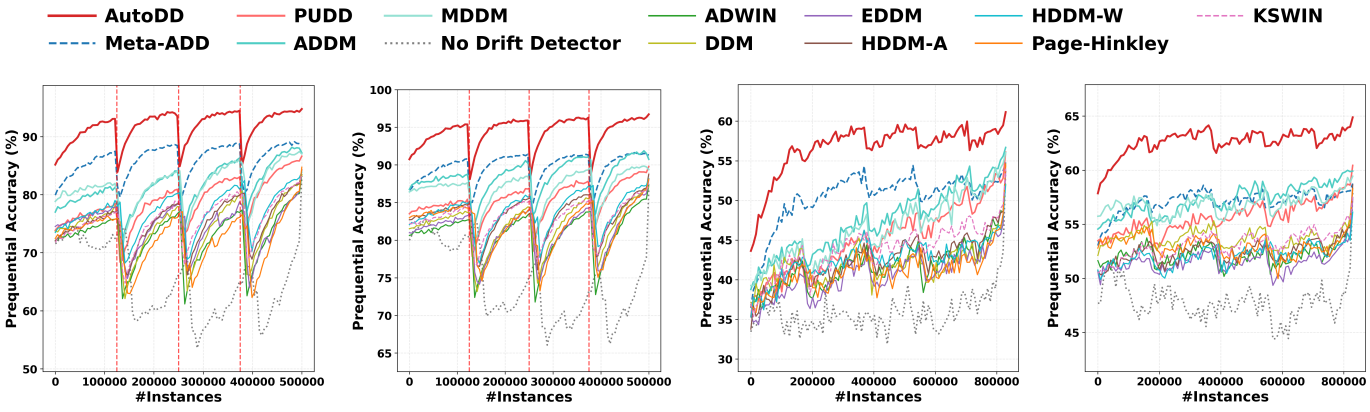
[1] Code: https://anonymous.4open.science/r/AutoDD-BCCF

TABLE IV: Results on synthetic datasets with both base models. **Bold**: best, Underline: second-best.

	GaussianNB (GNB)												Hoeffding Adaptive Tree (HAT)											
	SEA		HYP		AGR		RBF		RTG		Mean		SEA		HYP		AGR		RBF		RTG		Mean	
	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1
No Detector	83.8	0.40	76.4	0.34	78.8	0.41	50.4	0.09	58.2	0.10	69.5	0.27	86.2	0.42	79.1	0.36	81.3	0.43	53.1	0.11	60.8	0.12	72.1	0.29
ADWIN	84.0	0.61	81.9	0.00	71.5	0.51	69.6	0.00	64.5	0.00	74.3	0.22	88.1	0.64	85.5	0.06	74.6	0.55	73.3	0.07	68.3	0.11	77.9	0.29
DDM	83.8	0.00	86.1	0.63	71.6	0.49	71.3	0.64	64.4	0.00	75.4	0.35	88.0	0.10	89.0	0.67	74.7	0.52	74.6	0.69	68.1	0.13	78.9	0.42
EDDM	84.3	0.40	85.7	0.63	65.3	0.24	67.7	0.38	65.0	0.40	73.6	0.41	88.0	0.46	88.6	0.68	69.2	0.29	71.8	0.42	68.8	0.47	77.3	0.46
HDDM-A	84.6	0.69	81.5	0.00	72.2	0.63	69.7	0.00	64.4	0.00	74.5	0.26	88.3	0.64	86.8	0.06	75.1	0.63	73.8	0.07	69.1	0.11	78.6	0.30
HDDM-W	84.4	0.66	86.6	0.71	73.1	0.67	70.1	0.51	65.8	0.56	76.0	0.62	88.4	0.65	89.5	0.72	75.9	0.67	73.9	0.53	69.7	0.58	79.5	0.63
Page-Hinkley	84.7	0.70	81.5	0.00	71.5	0.49	69.4	0.00	64.5	0.00	74.3	0.24	88.2	0.65	86.0	0.06	74.8	0.52	73.5	0.07	68.5	0.11	78.2	0.28
KSWIN	83.7	0.35	85.7	0.65	71.5	0.50	70.0	0.61	65.2	0.39	75.2	0.50	88.0	0.42	88.3	0.66	74.5	0.53	73.7	0.63	68.9	0.43	78.7	0.53
MDDM	86.7	0.65	87.2	0.68	72.3	0.71	69.8	0.56	66.1	0.56	76.4	0.63	89.1	0.67	90.3	0.75	75.5	0.75	73.8	0.60	69.8	0.61	78.4	0.68
ADDM	87.7	0.83	88.7	0.73	73.6	0.67	70.9	0.62	67.6	0.71	77.7	0.71	89.8	0.86	91.9	0.77	76.6	0.72	75.0	0.69	71.3	0.77	79.3	0.76
PUDD	86.2	0.81	89.5	0.84	73.7	0.72	73.4	0.72	67.8	0.71	78.1	0.76	89.8	0.83	92.6	0.83	77.1	0.76	76.4	0.74	71.9	0.74	79.7	0.78
Meta-ADD	88.9	0.87	90.8	**0.91**	74.1	0.79	**78.0**	0.70	**72.9**	0.76	80.9	0.81	90.9	0.91	92.5	0.92	77.1	0.81	**84.8**	0.74	**80.4**	0.79	81.4	0.83
AutoDD	**93.1**	**0.90**	**94.6**	**0.91**	**78.5**	**0.84**	74.3	**0.81**	69.2	**0.84**	**81.9**	**0.86**	**96.7**	**0.93**	**98.4**	**0.93**	**82.4**	**0.90**	75.8	**0.84**	70.9	**0.88**	**84.0**	**0.87**

TABLE V: Results on real-world datasets with both base models. DN = detected drifts. **Bold**: best, Underline: second-best.

	GaussianNB (GNB)												Hoeffding Adaptive Tree (HAT)											
	Elec		Weather		Spam		Airlines		Poker		Mean		Elec		Weather		Spam		Airlines		Poker		Mean	
	Acc	DN	Acc	DN	Acc	DN	Acc	DN	Acc	DN	Acc	DN	Acc	DN	Acc	DN	Acc	DN	Acc	DN	Acc	DN	Acc	DN
No Detector	73.2	0	68.9	0	77.2	0	59.3	0	54.6	0	66.6	0	76.5	0	71.4	0	80.1	0	62.1	0	56.9	0	69.4	0
ADWIN	79.2	22	69.3	7	92.1	2	62.5	76	54.3	3	71.5	22.0	82.4	22	73.1	9	94.6	2	66.3	76	57.7	3	74.8	22.4
DDM	82.3	22	70.6	5	91.8	2	61.8	20	54.2	3	72.1	10.4	85.5	22	73.5	7	93.9	2	65.2	20	58.0	3	75.2	10.8
EDDM	82.4	22	73.8	9	92.9	2	61.3	182	53.6	4	72.8	43.8	85.2	22	77.1	9	94.8	2	64.6	182	57.8	4	75.9	43.8
HDDM-A	82.7	22	72.2	9	92.5	2	62.6	114	54.6	6	72.9	30.6	86.3	22	75.8	9	94.7	2	66.1	114	58.5	6	76.3	30.6
HDDM-W	81.9	22	72.6	9	91.9	1	61.0	269	52.3	394	71.9	139.0	85.1	22	76.0	9	94.5	1	64.9	269	56.8	404	75.5	141.0
Page-Hinkley	78.0	16	69.4	6	91.8	1	62.8	64	54.3	17	71.3	20.8	81.7	16	72.8	6	94.2	1	66.5	64	57.5	17	74.5	20.8
KSWIN	81.1	22	68.0	9	92.6	2	61.6	269	53.7	266	71.4	113.6	84.3	22	71.2	9	94.9	2	65.4	269	57.0	266	74.6	113.6
MDDM	83.9	22	74.8	9	92.8	2	62.2	110	56.1	190	74.0	66.6	86.5	22	78.0	9	94.6	2	66.0	110	60.0	190	77.0	66.6
ADDM	84.8	22	76.3	9	93.9	2	63.9	74	56.5	177	75.1	56.8	87.3	22	79.2	9	95.8	2	66.9	74	59.6	177	77.8	56.8
PUDD	84.4	22	75.4	9	93.2	2	64.0	117	56.2	81	74.6	46.2	87.5	22	78.6	9	95.6	2	67.7	117	59.7	81	77.8	46.2
Meta-ADD	85.4	22	74.9	9	94.5	2	63.8	139	55.5	182	74.8	70.8	87.9	22	78.2	9	95.9	2	66.5	139	59.6	182	77.6	70.8
AutoDD	**89.6**	22	**80.0**	9	**97.9**	2	**68.2**	118	**59.7**	188	**79.1**	67.8	**93.8**	22	**85.1**	9	**99.0**	2	**72.6**	118	**65.1**	188	**83.1**	67.8

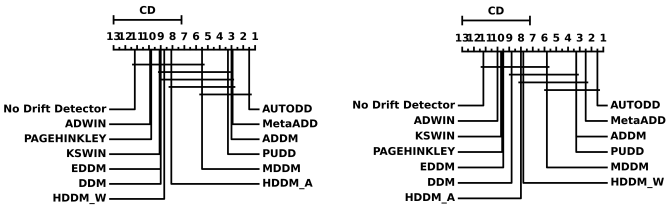


(a) SEA (GNB) (b) SEA (HAT) (c) Poker (GNB) (d) Poker (HAT)

Fig. 3: Prequential accuracy over time. SEA (synthetic) and Poker (real-world) datasets with GaussianNB and HAT base models.

C. Statistical Significance

To validate the observed performance differences, we conduct statistical significance analysis using the Friedman test with Nemenyi post-hoc analysis at $\alpha = 0.05$. Figures 4a and 4b present the Critical Difference diagrams for GaussianNB and HAT, respectively. AutoDD achieves the best average rank under both base models, with Meta-ADD, ADDM, PUDD, and MDDM following in subsequent ranks. The No Detector baseline performs worst, supporting the value of drift detection in these streaming benchmarks.



(a) GaussianNB (b) HAT

Fig. 4: Critical Difference diagrams (Friedman-Nemenyi, $\alpha = 0.05$). AutoDD achieves the best average rank under both base models.

D. Ablation Study

To understand the contribution of each component, we systematically remove AutoDD components and evaluate the impact on performance using GaussianNB as the base model. Table VI presents results averaged across all ten datasets. The hierarchical selection mechanism is important for AutoDD's performance. Removing Level 1 LinUCB detector selection and replacing it with random selection causes the largest

TABLE VI: Ablation study results across all datasets.

Configuration	Accuracy	F1	ΔAcc
Full AutoDD	**80.5%**	**0.64**	–
w/o Temporal	74.2%	0.59	-6.4%
w/o Statistical	71.2%	0.57	-9.4%
w/o Stationarity	73.5%	0.59	-7.1%
w/o LinUCB (Level 1)	62.5%	0.50	-18.0%
w/o Thompson (Level 2)	75.3%	0.60	-5.2%

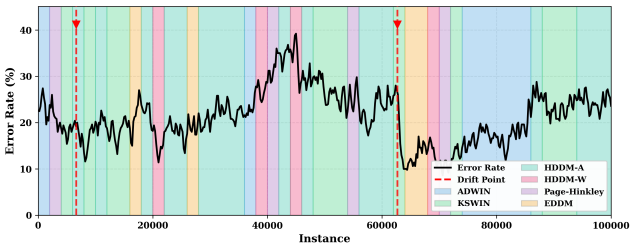
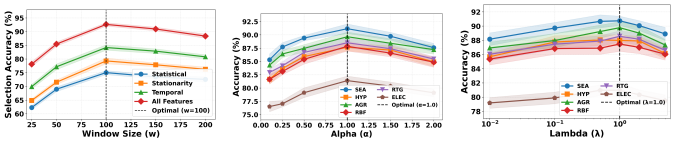


Fig. 5: AutoDD detector selection on the Hyperplane stream (single run, seed 42, w=100). Colored bands show the dominant LinUCB-selected detector per 2,000-instance block, the black curve is the windowed prequential error rate, and dashed red lines mark drifts flagged by the active detector.

performance drop, reducing accuracy from 80.5% to 62.5%, an 18.0% decrease. This degradation indicates that detector-stream matching is a major contributor to effective drift detection in our evaluation. Removing Level 2 Thompson Sampling hyperparameter configuration reduces accuracy to 75.3%, showing that adaptive tuning contributes beyond fixed default configurations. All three meta-feature categories contribute to the overall performance. Statistical features have the largest impact, with their removal reducing accuracy by 9.4%. Stationarity features contribute 7.1% to performance, while temporal features account for 6.4%. Figure 5 visualizes AutoDD's adaptive behavior on the Hyperplane stream (a single run, $w = 100$), showing how the LinUCB layer continually re-selects detectors as the stream's meta-features evolve rather than committing to a single detector, while the windowed prequential error stays bounded and recovers after detected drifts.

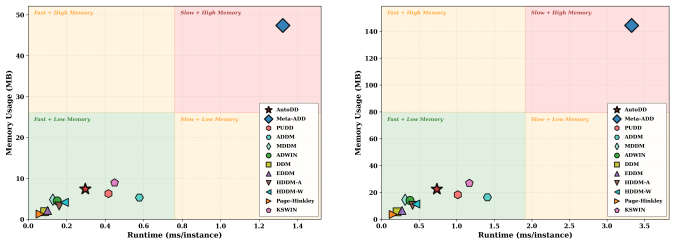
E. Sensitivity Study

We analyze AutoDD's sensitivity to key hyperparameters using GaussianNB as the base model (Figure 6). Among the tested values, window size $w = 100$ achieves the best selection accuracy, balancing statistical stability with drift responsiveness. Smaller windows suffer from high variance, while larger windows cause delayed drift response. For the exploration parameter, $\alpha_{\text{ucb}} = 1.0$ achieves the best performance among the tested values, with lower values causing exploitation bias and higher values leading to excessive exploration. The regularization parameter λ_{reg} shows robust performance across the range from 0.5 to 2.0, with $\lambda_{\text{reg}} = 1.0$ performing best in this sweep. This stability indicates that AutoDD is not overly sensitive to hyperparameter choices around the recommended defaults.



(a) w (b) α_{ucb} (c) λ_{reg}

Fig. 6: Hyperparameter sensitivity analysis.



(a) GaussianNB (b) HAT

Fig. 7: Runtime vs. memory trade-off. AutoDD lies in the Fast + Low Memory quadrant under both base models.

F. Computational Efficiency

Figures 7a and 7b compare runtime and memory usage across all methods for GaussianNB and HAT, respectively. AutoDD falls within the desirable *Fast + Low Memory* quadrant under both base models, demonstrating that its hierarchical selection mechanism introduces only modest overhead. Compared to state-of-the-art methods such as Meta-ADD and PUDD, AutoDD achieves competitive efficiency while delivering better detection performance, making it practical for high-velocity streaming scenarios.

G. Detection Delay

Beyond accuracy, timely drift detection is critical for rapid model adaptation. Table VII reports mean detection delay (in instances) across synthetic datasets with known drift points. AutoDD achieves the lowest mean delay of 38.4 instances, outperforming PUDD (41.8), Meta-ADD (44.7), ADDM (46.8), and all classical detectors in this comparison. This improvement is consistent with AutoDD selecting more responsive detectors in drift-prone regimes while avoiding over-sensitive detectors during stable periods.

H. Convergence Analysis

Table VIII shows how quickly AutoDD learns effective detector selections as the stream progresses, measured as the percentage of windows where the selected detector matches an offline proxy oracle under the same reward used for bandit feedback. AutoDD achieves 69.4% proxy-oracle selection accuracy in the first quarter of the stream, improving to 82.6% at mid-stream and 93.1% by stream end. This trend suggests that AutoDD learns increasingly reliable selections throughout the stream, while the metric should be interpreted as proxy-oracle agreement rather than direct proof of true drift-label optimality.

TABLE VII: Mean detection delay (instances) on synthetic datasets. Lower is better.

Method	SEA	HYP	AGR	RBF	Mean
ADWIN	45.2	78.3	52.1	81.4	64.3
DDM	38.7	62.5	48.9	65.2	53.8
EDDM	52.3	71.8	61.4	74.6	65.0
HDDM-A	43.8	75.1	50.3	78.6	61.9
HDDM-W	41.5	58.2	45.7	62.8	52.1
Page-Hinkley	46.9	80.5	53.8	83.2	66.1
KSWIN	49.1	68.7	58.2	71.4	61.9
MDDM	39.4	55.8	44.2	60.5	50.0
ADDM	36.5	51.3	41.6	57.8	46.8
PUDD	32.8	45.2	37.5	51.6	41.8
Meta-ADD	35.2	48.6	39.8	55.3	44.7
AutoDD	**28.4**	**42.1**	**34.2**	**48.7**	**38.4**

TABLE VIII: AutoDD proxy-oracle selection accuracy (%) at different stream positions.

Dataset	0–25%	25–50%	50–75%	75–100%
SEA	71.2	85.4	91.8	94.3
HYP	68.5	82.1	89.6	93.1
AGR	65.8	79.3	87.2	91.5
Elec	72.4	84.7	90.3	92.8
Poker	69.1	81.6	88.9	93.6
Mean	69.4	82.6	89.6	93.1

VI. Discussion and Limitations

We highlight several boundaries of the current design. First, AutoDD automates detector and hyperparameter selection but not the sliding-window size w, which the sensitivity analysis shows to be the most influential setting; $w = 100$ is robust across our benchmarks, yet selecting it for an unseen stream remains a manual step that an outer adaptive loop could close. Second, the reward is an accuracy-based proxy and therefore entangles detector quality with the downstream learner; we mitigate this through window-level smoothing and the significance threshold (Section III), but a fully label-free reward is an open problem. Third, to isolate detection from model selection we fix the base learner to GaussianNB and HAT, which are two standard but limited families, and adapt via full model reset, while richer learners and softer adaptation strategies (warm-starting, selective parameter resets, replay, ensemble updating) are left to future work. Finally, AutoDD targets tabular data streams, the prevalent setting in the drift-detection literature; non-tabular modalities such as image, audio, and text are outside the scope of this work. Within the tabular setting, the portfolio covers the principal detector families through their strongest and most recent representatives, so that the statistical two-sample test family such as IKS [34] and WRS [31] is represented by the newer KSWIN, HDDM, and ADWIN. Because AutoDD is detector-agnostic, incorporating additional tabular detectors is a natural extension of the framework rather than a change to its selection mechanism.

VII. Conclusion and Future Work

We propose AutoDD, a novel framework that automatically selects and configures drift detectors for data streams. The framework extracts comprehensive meta-features and employs a two-level hierarchical selection strategy using LinUCB for detector selection and Thompson Sampling for hyperparameter configuration. Under the stated proxy-reward and linearity assumptions, the theoretical analysis gives sublinear proxy-regret bounds for the selection layer. Extensive experimental results on synthetic and real-world datasets show that AutoDD achieves considerable improvements over fixed detector baselines and the compared state-of-the-art methods while requiring no expert intervention. Beyond the limitation-driven extensions discussed in Section VI, a further direction is multi-objective selection that jointly balances predictive accuracy, detection latency, and computational efficiency.

References

[1] J. ao Gama, I. Žliobaitė, A. Bifet, M. Pechenizkiy, and A. Bouchachia, "A survey on concept drift adaptation," *ACM Computing Surveys (CSUR)*, vol. 46, no. 4, pp. 1–37, 2014.

[2] V. Losing, B. Hammer, and H. Wersing, "Incremental on-line learning: A review and comparison of state of the art algorithms," *Neurocomputing*, vol. 275, pp. 1261–1274, 2018.

[3] J. Lu, A. Liu, F. Dong, F. Gu, J. Gama, and G. Zhang, "Learning under concept drift: A review," *IEEE Transactions on Knowledge and Data Engineering*, vol. 31, no. 12, pp. 2346–2363, 2019.

[4] J. Gama, P. Medas, G. Castillo, and P. Rodrigues, "Learning with drift detection," in *Brazilian Symposium on Artificial Intelligence*. Springer, 2004, pp. 286–295.

[5] A. Bifet and R. Gavaldà, "Learning from time-changing data with adaptive windowing," in *Proceedings of the 2007 SIAM International Conference on Data Mining*. SIAM, 2007, pp. 443–448.

[6] M. Z. A. Mayaki and M. Riveill, "Autoregressive based drift detection method," in *Proceedings of the International Joint Conference on Neural Networks (IJCNN)*. IEEE, 2022.

[7] C. Raab, M. Heusinger, and F.-M. Schleif, "Reactive soft prototype computing for concept drift streams," *Neurocomputing*, vol. 416, pp. 340–351, 2020.

[8] P. Lu, J. Lu, A. Liu, and G. Zhang, "Early concept drift detection via prediction uncertainty," in *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2025.

[9] H. Yu, J. Lu, and G. Zhang, "Meta-ADD: A meta-learning based pre-trained model for concept drift active detection," *Information Sciences*, vol. 578, pp. 1–14, 2021.

[10] F. Hutter, L. Kotthoff, and J. Vanschoren, "Automated machine learning: Methods, systems, challenges," in *Automated Machine Learning*. Springer, 2019.

[11] B. Celik, P. Singh, and J. Vanschoren, "Online AutoML: An adaptive AutoML framework for online learning," *Machine Learning*, vol. 112, no. 6, pp. 1897–1921, 2023.

[12] N. Verma, A. Bifet, B. Pfahringer, and M. Bahri, "ASML: A scalable and efficient AutoML solution for data streams," in *Proceedings of the Third International Conference on Automated Machine Learning (AutoML)*, ser. Proceedings of Machine Learning Research. PMLR, 2024.

[13] L. Li, W. Chu, J. Langford, and R. E. Schapire, "A contextual-bandit approach to personalized news article recommendation," in *Proceedings of the 19th International Conference on World Wide Web (WWW)*. ACM, 2010, pp. 661–670.

[14] S. Agrawal and N. Goyal, "Analysis of Thompson sampling for the multi-armed bandit problem," in *Proceedings of the 25th Annual Conference on Learning Theory (COLT)*. PMLR, 2012, pp. 39.1–39.26.

[15] A. Bifet, G. Holmes, B. Pfahringer, P. Kranen, H. Kremer, T. Jansen, and T. Seidl, "MOA: Massive online analysis, a framework for stream classification and clustering," in *Proceedings of the First Workshop on Applications of Pattern Analysis*. PMLR, 2010, pp. 44–50.

[16] M. Baena-García, J. del Campo-Ávila, R. Fidalgo, A. Bifet, R. Gavaldà, and R. Morales-Bueno, "Early drift detection method," in *Fourth International Workshop on Knowledge Discovery from Data Streams*, vol. 6, 2006, pp. 77–86.
[17] I. Friás-Blanco, J. de Campo-Ávila, G. Ramos-Jiménez, R. Morales-Bueno, A. Ortiz-Díaz, and Y. Caballero-Mota, "Online and non-parametric drift detection methods based on Hoeffding's bounds," *IEEE Transactions on Knowledge and Data Engineering*, vol. 27, no. 3, pp. 810–823, 2015.
[18] E. S. Page, "Continuous inspection schemes," *Biometrika*, vol. 41, no. 1/2, pp. 100–115, 1954.
[19] A. Pesaranghader, H. Viktor, and E. Paquet, "McDiarmid drift detection methods for evolving data streams," in *Proceedings of the International Joint Conference on Neural Networks (IJCNN)*. IEEE, 2017, pp. 1663–1670.
[20] R. S. Tsay, "Testing and modeling threshold autoregressive processes," *Journal of the American Statistical Association*, vol. 84, pp. 231–240, 1989. [Online]. Available: https://api.semanticscholar.org/CorpusID:17630406
[21] D. Kifer, S. Ben-David, and J. Gehrke, "Detecting change in data streams," in *Proceedings of the 30th International Conference on Very Large Data Bases (VLDB)*, vol. 4, 2004, pp. 180–191.
[22] T. Dasu, S. Krishnan, S. Venkatasubramanian, and K. Yi, "An information-theoretic approach to detecting changes in multidimensional data streams," in *Proceedings of Symposium on the Interface of Statistics, Computing Science, and Applications*, 2006.
[23] N. Lu, G. Zhang, and J. Lu, "Concept drift detection via competence models," *Artificial Intelligence*, vol. 209, pp. 11–28, 2014.
[24] V. M. A. Souza, A. R. S. Parmezan, F. A. Chowdhury, and A. Mueen, "Efficient unsupervised drift detector for fast and high-dimensional data streams," *Knowledge and Information Systems*, vol. 63, no. 6, pp. 1497–1527, 2021.
[25] C. Kulbach, J. Montiel, M. Bahri, M. Heyden, and A. Bifet, "Evolution-based online automated machine learning," in *Advances in Knowledge Discovery and Data Mining (PAKDD)*, ser. Lecture Notes in Computer Science, vol. 13280. Springer, Cham, 2022, pp. 472–484.
[26] Y. Abbasi-Yadkori, D. Pál, and C. Szepesvári, "Improved algorithms for linear stochastic bandits," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 24, 2011, pp. 2312–2320.
[27] T. Lattimore and C. Szepesvári, *Bandit Algorithms*. Cambridge University Press, 2020.
[28] V. Dani, T. P. Hayes, and S. M. Kakade, "Stochastic linear optimization under bandit feedback," in *Proceedings of the 21st Annual Conference on Learning Theory*, 2008, pp. 355–366.
[29] S. Agrawal and N. Goyal, "Further optimal regret bounds for Thompson sampling," in *Proceedings of the 16th International Conference on Artificial Intelligence and Statistics*. PMLR, 2013, pp. 99–107.
[30] P. Domingos and G. Hulten, "Mining high-speed data streams," in *Proceedings of the Sixth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. ACM Press, 2000, pp. 71–80.
[31] R. S. M. d. Barros and S. G. T. d. C. d. Santos, "A large-scale comparison of concept drift detectors," *Information Sciences*, vol. 451–452, pp. 348–370, 2018.
[32] V. M. A. Souza, D. M. dos Reis, A. G. Maletzke, and G. E. A. P. A. Batista, "Challenges in benchmarking stream learning algorithms with real-world data," *Data Mining and Knowledge Discovery*, vol. 34, no. 6, pp. 1805–1858, 2020.
[33] J. Montiel, M. Halford, S. M. Mastelini, G. Bolmier, R. Sourty, R. Vaysse, A. Zouitine, H. M. Gomes, J. Read, T. Abdessalem, and A. Bifet, "River: Machine learning for streaming data in Python," *Journal of Machine Learning Research*, vol. 22, no. 110, pp. 1–8, 2021.
[34] D. M. dos Reis, P. Flach, S. Matwin, and G. Batista, "Fast unsupervised online drift detection using incremental Kolmogorov-Smirnov test," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2016, pp. 1545–1554.

APPENDIX

Per-Drift-Type Performance: Table IX shows AutoDD's performance per drift type. AutoDD excels on sudden (88.2%), incremental (80.4%), and recurring (79.6%) drift. Meta-ADD slightly outperforms on gradual drift (82.3% vs. 81.7%) due to its learned drift-type representations.

TABLE IX: Accuracy (%) by drift type (GaussianNB).

Method	Sudden	Gradual	Incremental	Recurring
ADWIN	82.4	71.3	68.5	75.2
DDM	79.8	74.6	72.1	71.8
EDDM	75.2	78.4	74.8	69.3
HDDM-A	81.7	72.1	69.8	74.5
HDDM-W	80.1	76.2	73.5	74.6
Page-Hinkley	81.9	70.8	68.2	75.8
KSWIN	76.5	77.8	75.2	72.1
MDDM	81.3	75.8	74.1	73.9
ADDM	82.9	77.2	75.6	74.8
PUDD	82.1	79.5	76.8	75.4
Meta-ADD	84.6	**82.3**	78.9	77.8
AutoDD	**88.2**	81.7	**80.4**	**79.6**

TABLE X: Detector selection frequency (%) by stream characteristic.

Detector	High Velocity	High Noise	High Dim.
ADWIN	28.4	12.3	15.7
DDM	22.1	8.5	11.2
EDDM	8.3	18.7	14.5
HDDM-A	15.2	10.4	12.8
HDDM-W	12.6	14.2	13.1
Page-Hinkley	5.8	8.1	7.2
KSWIN	7.6	27.8	25.5

Detector Selection Patterns: Table X shows that AutoDD's learned selection policy aligns with domain knowledge: ADWIN and DDM are preferred for high-velocity streams, while KSWIN dominates for high-noise and high-dimensional streams.

This appendix gives the full proofs of the regret results stated in the Theory section. The condensed arguments in the main text (Theorem 1, Proposition 2, and the Corollary) are sketches of the derivations below.

Preliminaries and notation. Let $\mathcal{S} = \{(\mathbf{x}_t, y_t)\}_{t=1}^{\infty}$ denote the data stream, $\mathcal{D} = \{D_1, \ldots, D_K\}$ with $K = 7$ the detector portfolio, and $\boldsymbol{\phi}_t \in \mathbb{R}^m$ the meta-feature vector at time t with $m = 15$. For LinUCB we maintain, for each detector i, the regularized design matrix $\mathbf{A}_{i,t} = \lambda_{\text{reg}}\mathbf{I} + \sum_{s \le t:\, i_s = i} \boldsymbol{\phi}_s \boldsymbol{\phi}_s^\top$ and the response vector $\mathbf{b}_{i,t} = \sum_{s \le t:\, i_s = i} r_s \boldsymbol{\phi}_s$, with ridge estimate $\hat{\boldsymbol{\theta}}_{i,t} = \mathbf{A}_{i,t}^{-1}\mathbf{b}_{i,t}$. We write $N_i(T) = |\{t \le T : i_t = i\}|$ for the number of rounds detector i is selected, so that $\sum_{i=1}^{K} N_i(T) = T$. Throughout, A1–A3 are the assumptions of the Theory section: $\|\boldsymbol{\phi}_t\|_2 \le L$ with $L = \sqrt{m}$, linear reward $\mathbb{E}[r_{i,t} \mid \boldsymbol{\phi}_t] = \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_i^*$ with $\|\boldsymbol{\theta}_i^*\|_2 \le S$, and σ_{noise}-sub-Gaussian noise. All bounds are stated with respect to the proxy reward of Eq. (7), as discussed under assumption A4.

A. Proof of Theorem 1 (Level 1 Proxy Regret)

Theorem 1. *Under assumptions A1–A3, with exploration parameter* $\alpha_{ucb} = \sigma_{noise}\sqrt{m \log(KT/\delta_{conf})} + S$, *LinUCB detector selection achieves with probability at least* $1 - \delta_{conf}$*:*

$$R_1(T) = \sum_{t=1}^{T} \left(\max_i \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_i^* - \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_{i_t}^* \right) = O(\sqrt{KmT \log T}). \tag{16}$$

Proof. The proof proceeds in three steps.

Step 1: Confidence ellipsoid construction. By the self-normalized martingale inequality [26], for any $\delta_{\text{conf}} > 0$ and any fixed detector i, with probability at least $1 - \delta_{\text{conf}}/K$, for all $t \geq 1$,

$$\|\hat{\boldsymbol{\theta}}_{i,t} - \boldsymbol{\theta}_i^*\|_{\mathbf{A}_{i,t}} \leq \sigma_{\text{noise}} \sqrt{2 \log\Big(\frac{\det(\mathbf{A}_{i,t})^{1/2} \det(\lambda_{\text{reg}}\mathbf{I})^{-1/2}}{\delta_{\text{conf}}/K}\Big)} + \sqrt{\lambda_{\text{reg}}}\, S. \tag{17}$$

Using the determinant–trace inequality $\det(\mathbf{A}_{i,t}) \leq \big(\frac{\operatorname{tr}(\mathbf{A}_{i,t})}{m}\big)^m \leq \big(\lambda_{\text{reg}} + \frac{tL^2}{m}\big)^m$ and a union bound over all K detectors, with probability at least $1 - \delta_{\text{conf}}$, for all $i \in [K]$ and $t \geq 1$,

$$\|\hat{\boldsymbol{\theta}}_{i,t} - \boldsymbol{\theta}_i^*\|_{\mathbf{A}_{i,t}} \leq \alpha_{\text{ucb},t}, \tag{18}$$

where, for our choice of α_{ucb},

$$\alpha_{\text{ucb},t} = \sigma_{\text{noise}} \sqrt{m \log\big(1 + \tfrac{tL^2}{\lambda_{\text{reg}} m}\big) + 2 \log \tfrac{K}{\delta_{\text{conf}}}} + \sqrt{\lambda_{\text{reg}}}\, S \leq \alpha_{\text{ucb}},$$

with the $/m$ in the log argument following from the determinant–trace bound above. Call this the *confidence event*; the remainder of the proof is conditioned on it.

Step 2: Instantaneous regret bound. Let $i^* = \arg\max_i \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_i^*$ be the optimal detector at time t, and let the UCB rule select $i_t = \arg\max_i \big(\boldsymbol{\phi}_t^\top \hat{\boldsymbol{\theta}}_{i,t} + \alpha_{\text{ucb}} \|\boldsymbol{\phi}_t\|_{\mathbf{A}_{i,t}^{-1}}\big)$. Under the confidence event,

$$\begin{aligned}
\boldsymbol{\phi}_t^\top \boldsymbol{\theta}_{i^*}^* &\leq \boldsymbol{\phi}_t^\top \hat{\boldsymbol{\theta}}_{i^*,t} + \alpha_{\text{ucb}} \|\boldsymbol{\phi}_t\|_{\mathbf{A}_{i^*,t}^{-1}} && (19)\\
&\leq \boldsymbol{\phi}_t^\top \hat{\boldsymbol{\theta}}_{i_t,t} + \alpha_{\text{ucb}} \|\boldsymbol{\phi}_t\|_{\mathbf{A}_{i_t,t}^{-1}} && (20)\\
&\leq \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_{i_t}^* + 2\alpha_{\text{ucb}} \|\boldsymbol{\phi}_t\|_{\mathbf{A}_{i_t,t}^{-1}}, && (21)
\end{aligned}$$

where the first and third inequalities use the confidence bound and the second uses the UCB selection rule. Hence the instantaneous regret obeys

$$r_t^{(1)} = \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_{i^*}^* - \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_{i_t}^* \leq 2\alpha_{\text{ucb}} \|\boldsymbol{\phi}_t\|_{\mathbf{A}_{i_t,t}^{-1}}. \tag{22}$$

Step 3: Cumulative regret via the elliptical potential lemma and the $\sqrt{K}$ factor. Because the portfolio uses disjoint per-detector estimators, we split the cumulative regret by the played detector and bound each subsequence separately. Fix a detector i and consider the rounds $\{t : i_t = i\}$ on which it is played. By the elliptical potential lemma [28] applied to this subsequence (in its $\min(1, \cdot)$ form, so the bound holds without requiring $\lambda_{\text{reg}} \geq L^2$) and Cauchy–Schwarz,

$$\begin{aligned}
\sum_{t:\, i_t = i} \|\boldsymbol{\phi}_t\|_{\mathbf{A}_{i,t}^{-1}} &\leq \sqrt{N_i(T) \sum_{t:\, i_t = i} \|\boldsymbol{\phi}_t\|^2_{\mathbf{A}_{i,t}^{-1}}} && (23)\\
&\leq \sqrt{N_i(T)\, 2m \log\Big(1 + \tfrac{N_i(T) L^2}{\lambda_{\text{reg}} m}\Big)}. && (24)
\end{aligned}$$

Summing the instantaneous bound over all detectors and applying Cauchy–Schwarz across detectors, $\sum_{i=1}^K \sqrt{N_i(T)} \leq \sqrt{K \sum_{i=1}^K N_i(T)} = \sqrt{KT}$, gives

$$\begin{aligned}
R_1(T) = \sum_{i=1}^K \sum_{t:\, i_t = i} r_t^{(1)} &\leq 2\alpha_{\text{ucb}} \sum_{i=1}^K \sum_{t:\, i_t = i} \|\boldsymbol{\phi}_t\|_{\mathbf{A}_{i,t}^{-1}} && (25)\\
&\leq 2\alpha_{\text{ucb}} \sqrt{2m \log\Big(1 + \tfrac{TL^2}{\lambda_{\text{reg}} m}\Big)} \sum_{i=1}^K \sqrt{N_i(T)} && (26)\\
&\leq 2\alpha_{\text{ucb}} \sqrt{2mKT \log\Big(1 + \tfrac{TL^2}{\lambda_{\text{reg}} m}\Big)} && (27)\\
&= O(\sqrt{KmT \log T}), && (28)
\end{aligned}$$

where we used $\log(1 + N_i(T)L^2/(\lambda_{\text{reg}} m)) \leq \log(1 + TL^2/(\lambda_{\text{reg}} m))$ to pull the logarithmic factor out of the per-detector sum. The extra $\sqrt{K}$, relative to the single-estimator $O(\sqrt{mT \log T})$ bound, is exactly the price of maintaining K separate detector estimators and aggregating their elliptical potentials.

The last line follows the standard LinUCB convention of absorbing the slowly growing confidence radius $\alpha_{\text{ucb}} = \Theta(\sqrt{m \log T})$ into the $O(\cdot)$ constant. Retaining it gives the dimension-explicit rate

$$R_1(T) = O\big(\alpha_{\text{ucb}} \sqrt{KmT \log T}\big) = \tilde{O}\big(m\sqrt{KT}\big), \tag{29}$$

which is linear in m and matches the canonical OFUL bound $O(d\sqrt{T} \log T)$ [26] with $d = m$; the $\sqrt{KmT \log T}$ form quoted in Theorem 1 and used for the convergence estimate in the main text is its constant-radius shorthand. This completes the proof. □

B. Proof of Proposition 2 (Level 2 Proxy Regret)

Proposition 2. *For Thompson Sampling on C configurations per detector under the standard Bernoulli-sampling reduction for bounded rewards, with $N_i(T)$ selections of detector $i \in [K]$, the expected cumulative proxy regret satisfies*

$$R_2(T) = \sum_{i=1}^K O(\sqrt{C \cdot N_i(T) \log N_i(T)}). \tag{30}$$

Proof. We analyze the hyperparameter selection problem for each detector separately.

Step 1: Reduction to a multi-armed bandit. Conditioned on detector i being selected on the rounds $\{t : i_t = i\}$, hyperparameter selection reduces to a C-armed bandit over $N_i(T)$ rounds. The proxy rewards $r_t \in [-1, 1]$ are normalized to $\tilde{r}_t = (r_t + 1)/2 \in [0, 1]$.

Step 2: Beta–Bernoulli Thompson Sampling analysis. For bounded rewards in $[0, 1]$, the standard reduction draws a Bernoulli outcome with mean $\tilde{r}_t$ and updates a Beta posterior with that sampled outcome. By the analysis of [14], Thompson Sampling with $\text{Beta}(1, 1)$ priors on a C-armed bandit attains the gap-dependent expected regret

$$\mathbb{E}[R_i(N_i)] \leq O\left(\sum_{c:\, \Delta_c > 0} \frac{\log N_i}{\Delta_c}\right), \tag{31}$$

where $\Delta_c = \mu^* - \mu_c$ is the suboptimality gap of configuration c. The corresponding problem-independent (gap-free) bound, Theorem 2 of [29], is

$$\mathbb{E}[R_i(N_i)] \leq O(\sqrt{C \cdot N_i(T) \log N_i(T)}). \tag{32}$$

The fractional Beta update used in implementation (Eq. (5)) is the conditional expectation of this Bernoulli reduction and serves as a stable implementation heuristic rather than a separate regret theorem.

Step 3: Aggregation across detectors. Because the detector selections partition the horizon ($\sum_{i=1}^{K} N_i(T) = T$) and each detector's hyperparameter problem is an independent bandit, the total Level 2 proxy regret is

$$R_2(T) = \sum_{i=1}^{K} \mathbb{E}[R_i] = \sum_{i=1}^{K} O(\sqrt{C \cdot N_i(T) \log N_i(T)}). \tag{33}$$

This completes the proof. □

C. Proof of the Corollary (Combined Proxy Regret)

Corollary. *Under the decomposition $R(T) = R_1(T) + R_2(T)$, where R_1 captures detector-selection suboptimality and R_2 configuration-selection suboptimality, the combined proxy regret is*

$$R(T) = O(\sqrt{KmT \log T} + \sqrt{CKT \log T}). \tag{34}$$

Proof. We combine the two bounds. From Theorem 1, $R_1(T) = O(\sqrt{KmT \log T})$. For $R_2(T)$, applying $\log N_i(T) \leq \log T$ and Cauchy–Schwarz across detectors,

$$\begin{aligned} R_2(T) &= \sum_{i=1}^{K} O(\sqrt{C \cdot N_i(T) \log N_i(T)}) && (35)\\ &\leq O\left(\sqrt{C \log T} \sum_{i=1}^{K} \sqrt{N_i(T)}\right) && (36)\\ &\leq O\left(\sqrt{C \log T} \sqrt{K \textstyle\sum_{i=1}^{K} N_i(T)}\right) && (37)\\ &= O(\sqrt{CKT \log T}), && (38) \end{aligned}$$

using $\sum_i \sqrt{N_i(T)} \leq \sqrt{K \sum_i N_i(T)}$ and $\sum_i N_i(T) = T$. Adding the two terms yields $R(T) = O(\sqrt{KmT \log T} + \sqrt{CKT \log T})$. As in Theorem 1, the R_1 term is quoted in its constant-radius shorthand; carrying α_{ucb} replaces $\sqrt{KmT \log T}$ by $\tilde{O}(m\sqrt{KT})$, whereas the R_2 term needs no such convention since the Thompson-sampling constant is universal. Since both terms are sublinear in T, the average proxy regret satisfies $\lim_{T\to\infty} R(T)/T = 0$. This completes the proof. □